

JRAHS Year 12 Term 1 Mathematics (2U) 2008

Question 1

MARKS

a) Find: i) $\int (x+2)(x-7) dx$.

2

ii) $\int (5-8x)^9 dx$.

1

b) Evaluate: i) $\int_0^1 \frac{1}{e^{2x}} dx$.

2

ii) $\int_0^3 \frac{2x-1}{x+4} dx$.

3

c) On the same diagram shade the region where $y < \sqrt{4-x^2}$ and $y \geq x+2$ hold simultaneously.

3

d) i) Show $\frac{d}{dx}(x\sqrt{2x-4}) = \frac{x}{\sqrt{2x-4}} + \sqrt{2x-4}$.

1

ii) Hence find $\int \frac{x}{\sqrt{2x-4}} dx$.

3

Question 2 (Start a new page)

a)

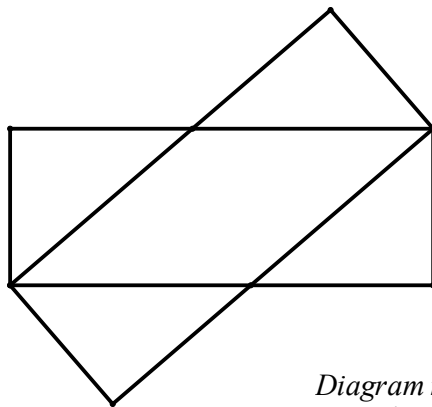


Diagram not

In the above diagram, $ABCD$ and $DEBF$ are two congruent rectangles with sides 3 and 7 units as in the diagram. ($AB = DF = 7$, $AD = DE = 3$).

Copy the diagram onto your writing booklet.

i) Prove that $\triangle ATD \equiv \triangle FTB$.

2

ii) Show that $AT = \frac{20}{7}$.

2

Question 2 (continued)**MARKS**

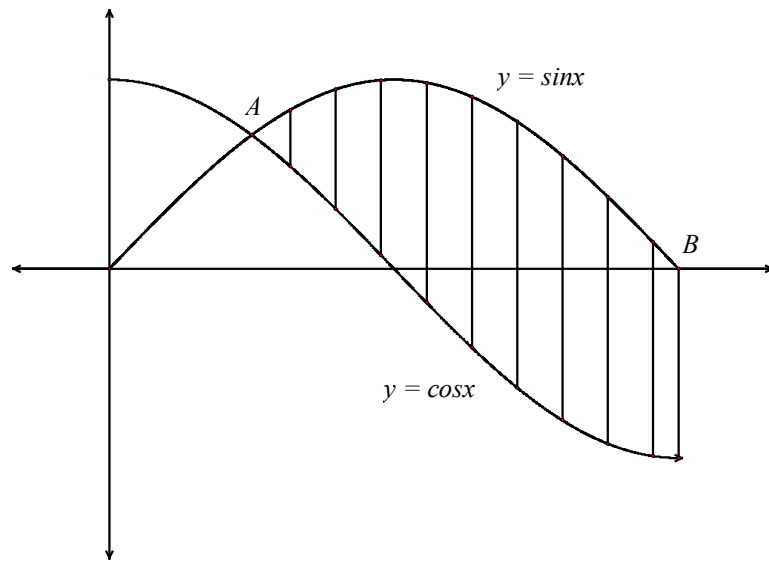
- b) Without using calculus, sketch $y = \frac{x+2}{x-3}$, showing all the essential features.

3

- c) Simplify $\sum_{k=1}^n \log\left(\frac{k+1}{k}\right)$.

3

d)



The diagram shows parts of the curves $y = \sin x$ and $y = \cos x$.

- i) Find the x -coordinates of the two points A and B .

2

- ii) Calculate the area of the shaded region.

3**Question 3 (Start a new page)**

- a) Find $\int \frac{e^{\tan x}}{\cos^2 x} dx$.

2

- b) The area bounded by curve $y = e^{2x}$, the y -axis and the line $y = 3$ is rotated about the y -axis to give a solid.

- i) Show that the volume V_y units³ of the solid formed, is given by

2

$$V_y = \frac{\pi}{4} \int_1^3 (\ln y)^2 dy.$$

- ii) Use Simpson's rule with 5 function values to find the volume of this solid, correct to 2 decimal places.

3

Question 3 (continued)**MARKS**

c) Find the area bounded by the graph of $y = x(x-1)(x-2)$ and the x -axis.

3

d) Consider the geometric series $1 + (3x-2) + (3x-2)^2 + \dots$

i) For what values of x does this series have a limiting sum?

3

ii) Find the value of x if this limiting sum has a value of $\frac{2}{3}$.

2**Question 4 (Start a new page)**

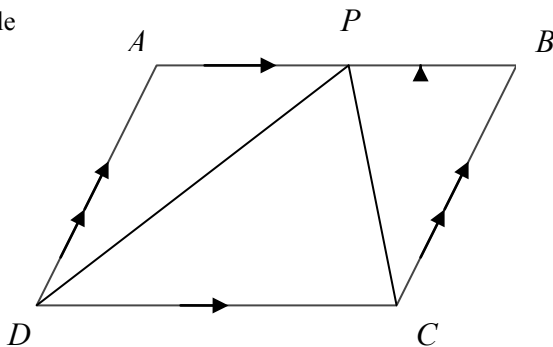
a) $ABCD$ is a parallelogram and P is on AB such that PD bisects $\angle ADC$ and PC bisects $\angle BCD$.

Copy the diagram onto your writing booklet.

Prove that $AB = 2 \times AD$.

4

Diagram not
to scale



Question 4 (continued)**MARKS**

- b) With the drought worsening, Ming has designed a counting generator that can simulate the number of rain drops per minute that fall over a pond during a storm.

The rain drops falling per minute forms the sequence:

$$\{1, 1, 3, 9, 23, \dots\}$$

with the n th term given by the formula $R_n = 1 - 2n + 2^n$.

- i) Verify that 115 is a term of this sequence. **1**
- ii) Find the total number of rain drops which fall over the pond in the first twenty-five minutes. **2**
- iii) If the surface area of the pond is 250m^2 find the average number of rain drops per cm^2 over the first twenty five minutes. **1**
- c) Katie has planned a holiday which she decides to take in 3 years time. She has estimated that the holiday will cost about \$8000 and plans to save a fixed amount each month for 36 months. She invests her savings at the beginning of each month in an account which pays interest at 6% p.a. compounded monthly at the end of each month.
- i) Let the amount she saves each month be \$ A and let \$ V_n be the value of her investment after n months. Show that the value of her investment at the end of 3 months is given by **1**
- $$V_n = A(1.005 + 1.005^2 + 1.005^3)$$
- ii) Find correct to the nearest dollar, the least amount of money that Katie would need to save each month to reach her target. **3**
- iii) If, after 2 years of her saving plan, the interest rate rose to 9% p.a., how much extra spending money would Katie have if she maintained the amount she was saving as calculated in part(ii) above? **3**

END

Q1a) $\int x^2 - 5x - 14 dx$

$= \frac{x^3}{3} - \frac{5x^2}{2} - 14x + c$ #

i) $\frac{(5-8x)^{10}}{-8 \times 10} + c$

$= \frac{(5-8x)^{10}}{-80} + c$ /

b) $= \left[\frac{e^{-2x}}{-2} \right]_0^1 = -\frac{1}{2}(1 + e^{-2}) = \frac{1}{2}(e^{-2} - 1)$ / #

ii) $\int_0^3 \frac{2(x+4)}{x+4} - \frac{9}{x+4} dx$

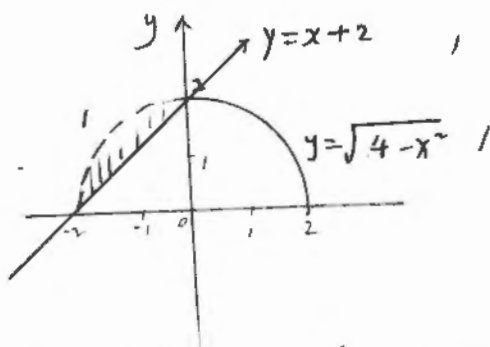
$= \int_0^3 2 - \frac{9}{x+4} dx$ /

$= \left[2x - 9 \ln(x+4) \right]_0^3$ /

$= 6 - 9 \ln 7 - 0 + 9 \ln 4$

$= 6 - 9 \ln \frac{7}{4}$ # /

c)



d) $\frac{d}{dx}(x\sqrt{2x-4}) = \frac{x \times 2}{\sqrt{2x-4}} + \sqrt{2x-4} \times 1$

$= \frac{x}{\sqrt{2x-4}} + \sqrt{2x-4}$ #

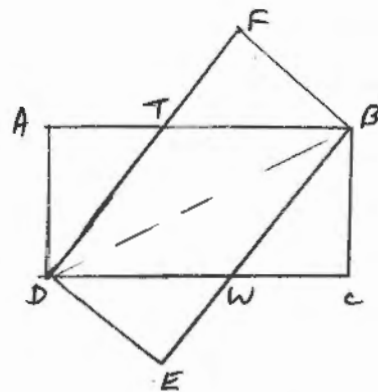
ii) FROM part (i)

$\int \frac{x dx}{\sqrt{2x-4}} = x \sqrt{2x-4} - \int \sqrt{2x-4} dx$ /

$= x \sqrt{2x-4} - \frac{2}{3} \frac{(2x-4)^{3/2}}{2} + c$ /

$= x \sqrt{2x-4} - \frac{(2x-4)^{3/2}}{3} + c$ /

Q2



i) ABCD and DEBF are both rectangles

$\angle BAD = \angle FDB = 90^\circ$ (angle of rectangle) $\frac{1}{2}$

$\angle ATD = \angle BTF$ (vertically opposite angles) $\frac{1}{2}$

$DE = FB$ (opposite sides of rectangle)

$AD = DE = 3$ (given)

$\therefore AD = FB$ $\frac{1}{2}$

$\therefore \triangle ATD \equiv \triangle BTF$ (AAS) $\frac{1}{2}$

or alternatively:

ABCD and DEBF are congruent rectangles

$\triangle ADB \equiv \triangle DFB$ (both equal half the congruent rectangles)

$\triangle ADB - \triangle TDB \equiv \triangle DFB - \triangle TDB$ /

$\therefore \triangle ATD \equiv \triangle BTF$

ii) Let $AT = x$ $TB = 7 - x$

$TF = x$ (corresponding sides of congruent triangles ATD, BTF)

By Pythagoras Th. $(7-x)^2 = x^2 + 3^2$ /

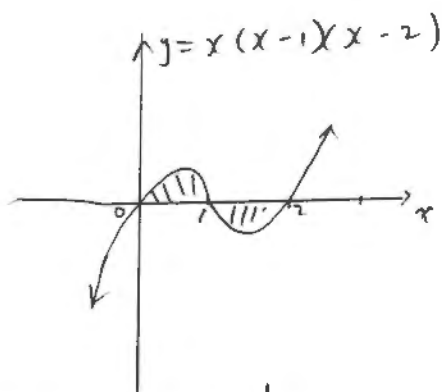
$49 - 14x + x^2 = x^2 + 9$

$14x = 49 - 9 = 40$

$x = \frac{20}{7}$ # /

Q3

c)



$$\text{Shaded Area} = \int_0^1 x(x-1)(x-2) dx +$$

$$\left| \int_1^2 x(x-1)(x-2) dx \right|$$

$$= \int_0^1 x^3 - 3x^2 + 2x dx + \int_1^2 x^3 - 3x^2 + 2x dx$$

$$= \left[\frac{x^4}{4} - x^3 + x^2 \right]_0^1 + \left[\frac{x^4}{4} - x^3 + x^2 \right]_1^2$$

$$= \left(\frac{1}{4} - 1 + 1 \right) + \left(\frac{16}{4} - 8 + 4 \right) - \left(\frac{1}{4} - 1 + 1 \right)$$

$$= \frac{1}{4} + \frac{1}{4}$$

$$= \frac{1}{2} \text{ unit}^2 \quad \#$$

$$s = \frac{1}{1-r}$$

$$d) \quad r = 3x-2$$

$$|3x-2| < 1$$

$$-1 < 3x-2 < 1$$

$$1 < 3x < 3$$

$$\frac{1}{3} < x < 1$$

$$\text{but } x \neq \frac{2}{3}$$

$$i) \quad \frac{2}{3} = \frac{1}{1-(3x-2)}$$

$$2 - (6x-4) = 3$$

$$6 - 6x = 3$$

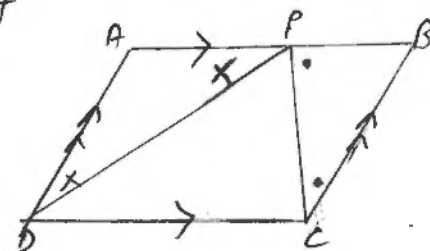
$$3 = 6x$$

$$x = \frac{1}{2}$$

#

Q4

a)

To prove $AB = 2x AD$.Proof : $AD \parallel BC$ (given) $\angle APD = \angle PDC$ (alternate angles, $AD \parallel PC$)

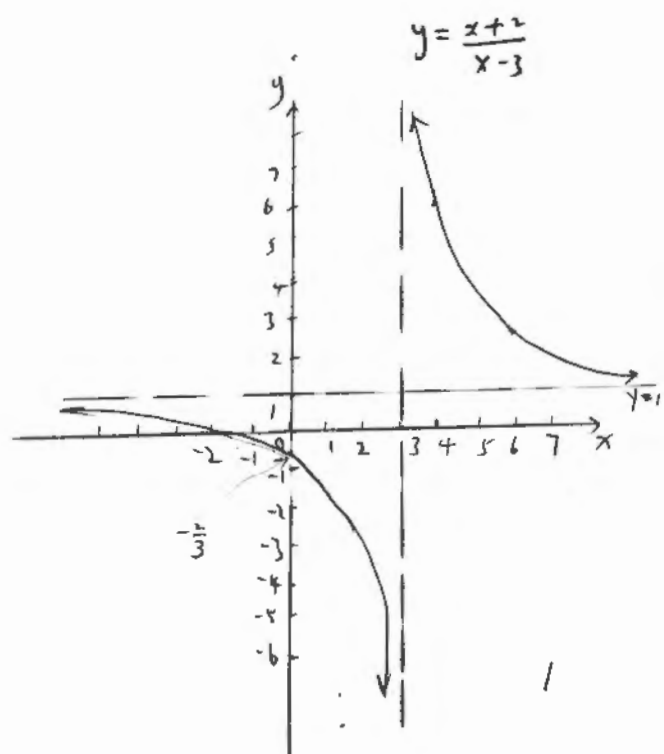
But $\angle ADP = \angle PDC$ (PD bisects $\angle ADC$)
 $\therefore AD = AP$ (sides opposite equal angles are equal)

Similarly $BC = BP$ But $AD = BC$ (opposites of para)

$$AD + BC = AP + BP$$

$$2AD = AB \quad \#$$

2b)



asymptotes $x=3$
 $y=1$

$x=0, y=-\frac{2}{3}$

$y=0, x=-2$

c) $\log \frac{2}{1} + \log \frac{3}{2} + \log \frac{4}{3} + \dots + \log \frac{n+1}{n}$

$= \log \frac{2}{1} \times \frac{3}{2} \times \frac{4}{3} \times \dots \times \frac{n}{n-1} \times \frac{n+1}{n}$

$= \log(n+1)$

d) $\sin x = \cos x$ when $x = \frac{\pi}{4}$

$\therefore x$ -coordinate of A = $\frac{\pi}{4}$

$\sin x = 0$ when $x = \pi$

x -coordinate of B = π

ii) Shaded Area = $\int_{\frac{\pi}{4}}^{\pi} \sin x - \cos x \, dx$
 $= [-\cos x - \sin x]_{\frac{\pi}{4}}^{\pi}$

$= -[\cos x + \sin x]_{\frac{\pi}{4}}^{\pi}$

$= -(-1 + 0) + (2 \times \frac{1}{\sqrt{2}})$

$= 1 + \frac{2}{\sqrt{2}} \text{ unit}^2$
 (or $1 + \sqrt{2} \text{ unit}^2$)

Q3

a) $\int e^{\tan x} \sec^2 x \, dx$

$= e^{\tan x} + c$

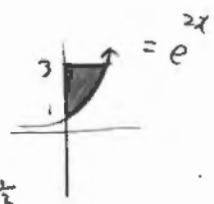
b) $y = e^{2x}$

$\ln y = 2x \therefore x = \frac{1}{2} \ln y$
 when $x=0, y=1$

Vol(V) = $\int_1^3 \pi x^2 \, dy$

$= \pi \int_1^3 \left(\frac{\ln y}{2}\right)^2 \, dy$

$= \frac{\pi}{4} \int_1^3 (\ln y)^2 \, dy$



ii)

y	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3
$(\ln y)^2$	0	0.1644	0.4805	0.8396	1.2069

Vol = $\frac{\pi}{4} \left(\frac{1}{6}\right) (0 + 4(0.1644 + 0.8396) + 2 \times 0.4805 + 1.2069)$

$= \frac{\pi}{24} (4.016 + 0.961 + 1.2069)$

$= 0.81 \text{ (2dp)}$

Q4

b) $R_n = 1 - 2n + 2^n$

i) $115 = 1 - 2n + 2^n$

When $n=7$

$$1 - 2 \cdot 7 + 2^7 = 1 - 14 + 128 = 115$$

$\therefore 115$ is a term in the series

ii) $\sum_{n=1}^{25} 1 - 2n + 2^n$

$$= 25 - \cancel{2} \left(\frac{1+25}{\cancel{2}} \right) + \frac{2(2^{25}-1)}{2-1}$$

$$= 25 - 26 \times 25 + 2(2^{25}-1)$$

$$= 67108237$$

iii) $250 \text{ m}^2 = 2500000 \text{ cm}^2$

$\therefore$ Ave No of rain drops

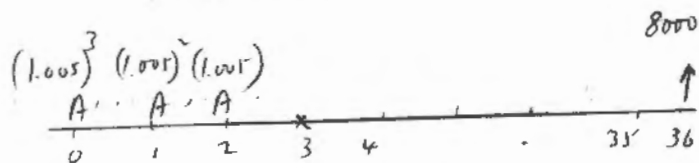
$$\text{per cm}^2 = \frac{67108237}{2500000} / \text{cm}^2$$

$$= 26.84 / \text{cm}^2$$

(2dp)

c) $6\% \text{ p.a.} = 0.5\% \text{ per month}$

$$1+i = 1.005$$



$$V_3 = A(1.005 + 1.005^2 + 1.005^3) \neq$$

ii) At the end of 36 months

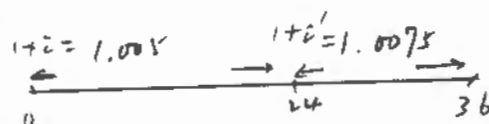
$$8000 \leq A(1.005 + 1.005^2 + \dots + 1.005^{36})$$

$$8000 \leq A \frac{1.005(1.005^{36}-1)}{1.005-1}$$

$$A \geq \$202.36 \text{ (2dp)}$$

$\therefore$ at least \$203 per month.

iii)



Savings at the end of 2 yrs (use $A = \$203$)

$$= 203(1.005) \left(\frac{1.005^{24}-1}{1.005-1} \right) = 5188.5003$$

Balance at the end of 3 yrs

$$= 203(1.0075) \left(\frac{1.0075^{12}-1}{1.0075-1} \right) +$$

$$5188.5003(1.0075)^{12}$$

$$= 2558.0828 + 5675.2174$$

$$= 8233.3002$$

Extra added value =

$$8233.3002 - 203 \times 1.005 \left(\frac{1.005^{36}-1}{1.005-1} \right)$$

$$= 8233.3002 - 8025.1555$$

$$= \underline{\underline{208.14 \text{ (2dp)}}}$$